A FIRST-PRINCIPLES GUIDE

Post-Labor Economics

What happens to the economy when machines can do all the mental and physical work

An argument built step by step from axioms,
six scenarios for 2026–2036, and a best guess

November 2026

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ABOUT THIS GUIDE

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The research notes, the data, the models and the code that draws every figure are in the repository:

https://github.com/lastmjs/post-labor-economics

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Nothing here is investment, legal or tax advice. It is an argument about the economy as a whole, and forecasts about the economy are often wrong.

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PART I

The machine we live inside

Before asking what breaks, we need to know how the thing works: what an economy is, how people get their share of it, and why two centuries of machines made workers richer and not poorer.

CHAPTER 1

What an economy is

An economy is a way of answering two questions: what gets made, and who gets it. Money, firms, jobs, banks and markets are machinery for answering those two questions. The premise of this guide, machines that can do all the work, makes the first question easier than it has ever been and throws the second one wide open.

To see why, we need a handful of facts about how the machinery works. This chapter sets out five axioms and draws three propositions from them. None of them is controversial among economists. The work lies in keeping all of them in view at once, because the argument of the later chapters is nothing more than these facts applied to a new situation.

Wants, production and ownership

> **AXIOM 1** **People want more than exists**
> At a price of zero, people would take more food, housing, medical care, travel, security and free time than can be supplied. Because wants exceed supply, every society needs some rule that decides who gets what.

Particular wants can be satisfied. Nobody wants a tenth meal in a day. But total wants have never been satisfied in any society on record. People who have enough food want better housing; people with good housing want health, schooling, travel and standing among their neighbors. Some wants cannot be satisfied for everyone even in principle, because they are about rank: the best plot of land, the most admired school, the front row. This matters later, because it means that no amount of production makes the question of distribution go away.

> **AXIOM 2** **Everything is made from four kinds of input**
> Every good and service is produced by combining:
> - **work**: human time, effort and skill;
> - **tools**: machines, buildings, vehicles and software, which are things made in order to make other things, and which economists call capital;
> - **nature**: land, energy and raw materials;
> - **knowledge**: the recipes for combining the other three.

The four inputs differ in one way that will turn out to decide almost everything. Tools can be manufactured: if more are wanted, more can be built. Knowledge, once found, can be used by everyone at once without running out. Nature is fixed in quantity or expands slowly. And work has been limited by the number of people and the hours in their days. An economy can double its stock of machines in a decade. It cannot double its stock of adults.

AXIOM 3 **Every input has an owner**

In a market economy, each input belongs to someone who decides how it is used and collects what it earns. A person's work belongs to that person. Nobody can be made to hand over what they own, except to the state through taxes.

DEFINITION 1 **Income**

A person's income is their claim on what the economy produces. In a market economy a claim can arise in only three ways: by **selling work** (wages), by **owning** something that earns a return (profits, interest, rent, royalties), or by **receiving a transfer** from someone who has a claim (family, charity or the state).

There is no fourth way. Every scheme discussed in this guide, from a universal basic income to a public wealth fund to a law that reserves certain jobs for humans, works through one of these three channels.

PROPOSITION 1 **All output is somebody's income**

Every dollar of output is paid to the owner of some input that helped produce it. The total income of a country therefore equals its total output, and the question "who gets the output?" is the same question as "who owns the inputs that production uses, and what is each input paid?"

Rests on: A2, A3

Follow the money from a single sale. A buyer pays $30,000 for a car. The carmaker pays some of it to its workers, some to its lenders, and some to suppliers of steel, glass and electricity; it keeps the remainder as profit, which belongs to its shareholders. Each supplier does the same with its part. Trace every branch to the end and the whole $30,000 has landed in someone's pocket as a wage, as interest, as rent or as profit, with the government taking a share along the way. Nothing is left over, and nothing is paid out that was not first earned by a sale. National statistics are built on this fact: gross domestic product and gross domestic income are two measurements of the same number.

This identity will do a great deal of work in Chapter 6. It means, for example, that wages cannot fall across an economy without the same dollars showing up as somebody else's income, so long as output itself does not fall.

What a price is

AXIOM 4 **Nobody pays more than they must**

A buyer who can get the same result in two ways takes the cheaper one. A firm that can have a task done by either of two methods of equal quality uses the method that costs less; a firm that does not is undersold by a competitor that does.

PROPOSITION 2 An input is paid for its scarcity, not its importance
The price of an input can never be more than the cost of the cheapest substitute that does the same job. Where there is no substitute, competition among buyers pushes the price up to the value of what one more unit of the input adds. In both cases, what sets the price is how hard it is to get one more unit, not how much the input matters in total.
Rests on: A4

This is the old puzzle of water and diamonds. Water is necessary for life and costs almost nothing. Diamonds are ornaments and cost a great deal. The puzzle dissolves once you notice that a price attaches to one more unit, not to the whole supply. One more gallon of water is easy to get; one more diamond is not.

Apply the same reasoning to work. A wage is the price of an hour of someone's work. It is high when what that hour adds cannot be had more cheaply in any other way, and it is low when it can. A wage is not a measure of effort, of merit or of how much the worker needs the money, and it is not a measure of how important the work is. Home health aides do work on which lives depend and are among the lowest-paid workers in the United States, because many people are able and willing to do that work. **A wage is a price, and it is governed by what governs every other price.** Keep hold of this sentence. Most confusion about automation comes from letting go of it.

How most people get their share

AXIOM 5 Most people own little except their capacity to work
In the United States the less wealthy half of households owns about 2.5 percent of all household wealth, and most of that is the house they live in. The wealthiest tenth owns about two thirds of all wealth and nearly nine tenths of all shares in companies. For a typical household, wages and the benefits that depend on wages are the main source of income for the whole of working life.

PROPOSITION 3 The wage system is the distribution system
Because most people own little except their work, most people's claim on output is their wage. The labor market is therefore doing two jobs at once. It is a **pricing** system, which tells firms where human work is worth using. And it is the **distribution** system, which decides how much of the output most households can buy.
Rests on: D1, P1, A5

The second job goes deeper than paychecks. In the United States most of the safety net is itself built on wages. Social Security and Medicare are financed by a tax on payrolls, and their benefits are earned through years of covered work. Unemployment insurance pays in proportion to past wages. About half the population gets health insurance through an employer. Mortgages are granted against expected wages. Even the federal government's own income rests mainly on work: taxes on wages and salaries supply most of its revenue.

The two jobs of the labor market have gone together for so long that they look like one thing. They are two. One is a matter of engineering and cost: is a person the cheapest way to get this task done? The other is a matter of how people live. For two hundred years the answer to the

first has been "yes, for a great many tasks," and so the second took care of itself. The rest of this guide asks what happens when the answer to the first question becomes "no."

WHERE THE CHAIN STANDS

- Output and income are the same total (**P1**). Whoever owns the inputs that production needs receives the output.
- Any input, human work included, is paid according to how hard it is to replace at the margin, and never more than its cheapest substitute costs (**P2**).
- Most people own only their work (**A5**), so the wage is how most people receive their share (**P3**).
- It follows that anything that makes human work easy to replace is a threat to distribution, even if it is a gift to production. To judge that threat we first have to understand why two centuries of machines did not carry it out.

CHAPTER 2

Why two centuries of machines made workers richer

Machines have been taking over human work since the 1700s, and over that time the pay of ordinary workers in the industrial countries has risen more than tenfold. Anyone who predicts that the next machines will be different has to explain first why the earlier ones were not. This chapter gives that explanation, and gives it in a form precise enough to test against the premise of this guide.

The standard reassurance goes like this. In 1800 about three quarters of Americans worked on farms. Today fewer than two in a hundred do, and the country grows far more food. The farm workers were not left idle forever; their grandchildren became machinists, clerks, nurses and programmers. People who warned that machines would cause permanent mass unemployment, from the Luddites in 1811 to the authors of the "Triple Revolution" memorandum sent to President Johnson in 1964, were wrong each time. Economists named the mistake the "lump of labor fallacy": the belief that there is a fixed amount of work to be done, so that a machine which does some of it leaves less for people.

The reassurance is correct about the past. But "it turned out fine before" is an observation, and what we need is the mechanism. A mechanism has conditions, and conditions can stop holding.

Machines take tasks, and tasks are complements

> DEFINITION 2 **Task**
> A task is one distinct piece of work needed to produce something: welding a seam, diagnosing a fault, answering a customer's question. A job is a bundle of tasks. A machine never takes over a job as such. It takes over tasks.

> AXIOM 6 **Most production needs many tasks, all of them**
> Making almost anything requires a number of different tasks, and more of one does not make up for the lack of another. A car needs designing, stamping, welding, wiring, testing, shipping and selling. Twice as much welding does not make up for no wiring. In the language of economics, tasks are *complements*.

The evidence for this axiom is all around. It is why a strike by a few hundred port crane operators can stop the trade of a whole coast. It is also the reason behind a famous pattern called Baumol's cost disease: the parts of the economy where productivity has grown slowly, such as health care and education, take a growing share of total spending, exactly because the things that became cheap could not be used in place of the things that did not.

Put the last two statements together with the rule about prices from the previous chapter and a result follows at once.

PROPOSITION 4 The step that machines cannot do collects the reward

When machines make some tasks cheap and plentiful, the remaining tasks become the scarce step in production. Since inputs are paid for scarcity, more of the value of the product flows to whoever performs the remaining tasks. If people are the only ones who can perform them, automation raises wages.

Rests on: P2, D2, A6

A WORKED EXAMPLE: THE SHIRT ECONOMY

A hundred workers make shirts. Each shirt needs one unit of cutting and one unit of sewing, and a worker can do one unit of either in an hour. Fifty workers cut and fifty sew, so the economy makes 50 shirts an hour. The workers are the only input, so they receive all 50 shirts between them: **half a shirt per hour of work**.

Now someone invents a machine that cuts cloth for next to nothing. All hundred workers move to sewing, and the economy makes 100 shirts an hour. Sewing is now the only scarce step, so the sewers collect nearly all the value: close to **one shirt per hour of work**. The machine took over half of all the tasks that workers used to do, and the workers' real pay doubled.

Notice what the result required. Sewing still had to be done by a person. The cutters had to be able to become sewers. And somebody had to want the extra 50 shirts. We will return to this economy in Chapter 4 and add a second machine.

The three conditions

The shirt example shows the mechanism in miniature, and it shows the conditions on which the mechanism depends. Stated generally:

PROPOSITION 5 Why past automation raised wages

Automation has raised the pay of workers when, and because, three conditions held together.

1. **Something was left.** There remained tasks that only people could do.
2. **What was left was needed.** Those tasks were complements to what the machines produced, so demand for them grew as machine output grew.
3. **People could get there.** Most workers, or their children, could move into the remaining tasks.

Rests on: P4, A1

For two hundred years the first condition held for a simple reason. Every machine ever built was *narrow*. A power loom weaves and does nothing else. A tractor pulls. A spreadsheet adds. A person, by contrast, is a general-purpose system: the same body and brain can weave, pull, add, negotiate, comfort a child and learn a trade that did not exist the year before. Whenever a narrow machine took over one task, there were always other tasks for which a general-purpose worker was the only candidate.

AXIOM 7 Until now, people could do things that no machine could
Through every earlier wave of automation, people kept abilities that no machine had: flexible reasoning, language, fine control of the hands in cluttered surroundings, and the understanding of other people. Each wave of machines pushed human work toward the abilities that remained exclusive.

You can read economic history as a sequence of such moves. Engines replaced muscle, and people moved to work that needed dexterity and attention on the factory line. Factory machines and then industrial robots took over the repetitive movements, and people moved to offices, to work that needed reading, calculating and coordinating. Computers took over the routine calculating and filing, and people moved again, toward work that needed judgment, persuasion, creativity and care, and toward manual work in unpredictable places, such as plumbing a hundred-year-old house. One study found that about 60 percent of American workers in 2018 held kinds of jobs that did not yet exist in 1940. The new jobs appeared, and people filled them, because people were the only general-purpose workers there were.

The second condition held because of **A6** and **A1**: the remaining tasks were needed, and there was no limit to what people wanted. The third held most of the time, though slowly and at real cost.

When the third condition failed

It is worth pausing on that cost, because history's reassurance is weaker than it sounds even for the episodes that ended well.

In Britain between about 1800 and 1840, output per worker rose by nearly half while real wages hardly moved. The gains went to the owners of the new factories. Only after about 1840 did wages begin to rise in step with productivity. The historian Robert Allen named these decades "Engels' pause," after Friedrich Engels, who described the condition of English workers in 1845 as the pause was ending. A worker who was twenty when it began was sixty when it ended.

The handloom weavers of England are the sharpest case. There were roughly a quarter of a million of them in the 1820s, skilled workers with a respected trade. The power loom did not make them unemployed at a stroke. It did something slower: it put a falling ceiling on what a piece of handwoven cloth could sell for. Their earnings fell by more than half over about three decades, and the trade had all but vanished by the 1860s. Few of the weavers themselves moved to anything better. The gains from the loom were real and eventually reached almost everyone, but they reached a different generation.

The more recent case is the arrival of Chinese manufactured imports in the United States after 2000. Economic theory predicted that workers in the affected towns would move to other industries or other places. Mostly they did not. More than a decade later the towns that had lost the most factory work still had lower employment, lower wages, more dependence on disability payments, fewer marriages and more deaths from drugs and alcohol.

The lesson to carry forward is that even when all three conditions hold for the economy as a whole, adjustment takes decades and its costs fall on particular people who are rarely compensated. That is the *good* case.

The horse

There is one large group of workers in modern history for whom the first condition failed completely. They were not human, which is why they are usually left out of the reassuring story, but the economics is exact.

In 1915 there were about 26 million horses and mules at work in the United States, pulling plows, wagons, streetcars and artillery. The horse had already survived one great wave of machines. Railways took over long-distance haulage in the 1800s and the number of working horses *rose*, because every ton that arrived by rail still had to be carted from the station. That was **P4** at work: the horse did the step the railway could not.

Then came the internal combustion engine, in the form of the tractor and the truck, and this machine was different in one respect. It could do everything a horse could do. No remaining step in farming or haulage needed a horse. From that point the price of a horse's work was capped by the cost of getting the same pulling and carrying from an engine (**P2**), and that cost fell every year. A horse, however, has an upkeep that does not fall: several acres' worth of feed, a stable, a farrier and a vet. Once the value of a horse's work dropped below the cost of keeping a horse, it no longer paid to breed one. By 1960 the country had about 3 million horses, and most of those were kept for pleasure.

> **PROPOSITION 6 The horse lemma**
> Suppose that for *every* task an input can perform there is a substitute that performs it at lower cost than the input's own upkeep. Then the price of the input falls below its upkeep, and its use in production falls toward zero. Total output rises while this happens.
> Rests on: **P2**

Nothing about this required horses to become worse at their work. A horse in 1950 was as strong and as willing as a horse in 1900. Nor did the economy have less need of pulling and carrying; it did far more of both. The work remained, and the horse stopped being the cheapest way to get it done.

The economist Wassily Leontief, who won the Nobel prize for mapping how industries feed into one another, drew the comparison in 1983: the role of humans as the most important input to production, he wrote, was bound to shrink in the same way the role of horses had. For forty years this was treated as a provocation, because the first condition of **P5** so obviously held. Machines were narrow.

Three ways people are not horses

The comparison is uncomfortable, and it should be handled with care, because there are three real differences between a population of horses and a population of people. Horses were never the purpose of the economy; people are. Horses owned nothing; people own things. And horses had no say in what happened to them; people vote.

Put in terms of this guide: people are the *customers* for whom production exists, they are (some of them) the *owners* of the machines, and they are the *citizens* of states that can tax and redistribute. Each of the three is a possible way out of the horse's fate. Whether any of them works is not settled by economics alone, and Parts II and III take them one at a time. For now it

is enough to state what the horse does establish: an input's long record of benefiting from new technology is no protection once a technology arrives that can do everything the input does.

WHERE THE CHAIN STANDS

- Machines take tasks. Because tasks are complements, the tasks that machines cannot do become the bottleneck and collect the reward (**P4**).
- Automation raised wages for two centuries because three conditions held: something was left for people to do, it was needed, and people could move into it (**P5**). The first condition held because machines were narrow and people were general (**A7**).
- Even then, adjustment took a generation or more and the losers were seldom made whole.
- Where the first condition fails entirely, the horse lemma applies (**P6**): the input's price falls below its upkeep while output grows.
- So the whole question turns on the first condition. The next chapter looks at a machine designed to remove it.

CHAPTER 3

The premise

CHAPTER 4

Wages

CHAPTER 5

Growth

CHAPTER 6

Distribution

CHAPTER 7

Residual

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Power

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Now

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